Data Mining

Week 4

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Predicting Fuel Efficiency

In the Week 4 Exercise, you will build a linear regression model to predict fuel efficiency (miles per gallon) of automobiles. Download the auto-mpg.csv dataset from: Auto-mpg dataset.

Load the data as a Pandas data frame and ensure that it imported correctly.

In [30]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

# Load the data
df = pd.read_csv('auto-mpg.csv')

# Inspect the data
print(df.head())
print(df.info())
```

```
    mpg  cylinders  displacement horsepower  weight  acceleration  model year  \
0  18.0          8         307.0        130    3504          12.0          70
1  15.0          8         350.0        165    3693          11.5          70
2  18.0          8         318.0        150    3436          11.0          70
3  16.0          8         304.0        150    3433          12.0          70
4  17.0          8         302.0        140    3449          10.5          70

   origin                   car name
0       1  chevrolet chevelle malibu
1       1          buick skylark 320
2       1         plymouth satellite
3       1              amc rebel sst
4       1                ford torino
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 398 entries, 0 to 397
Data columns (total 9 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   mpg           398 non-null    float64
 1   cylinders     398 non-null    int64
 2   displacement  398 non-null    float64
 3   horsepower    398 non-null    object
 4   weight        398 non-null    int64
 5   acceleration  398 non-null    float64
 6   model year    398 non-null    int64
 7   origin        398 non-null    int64
 8   car name      398 non-null    object
dtypes: float64(3), int64(4), object(2)
memory usage: 28.1+ KB
None
```

Begin by prepping the data for modeling:

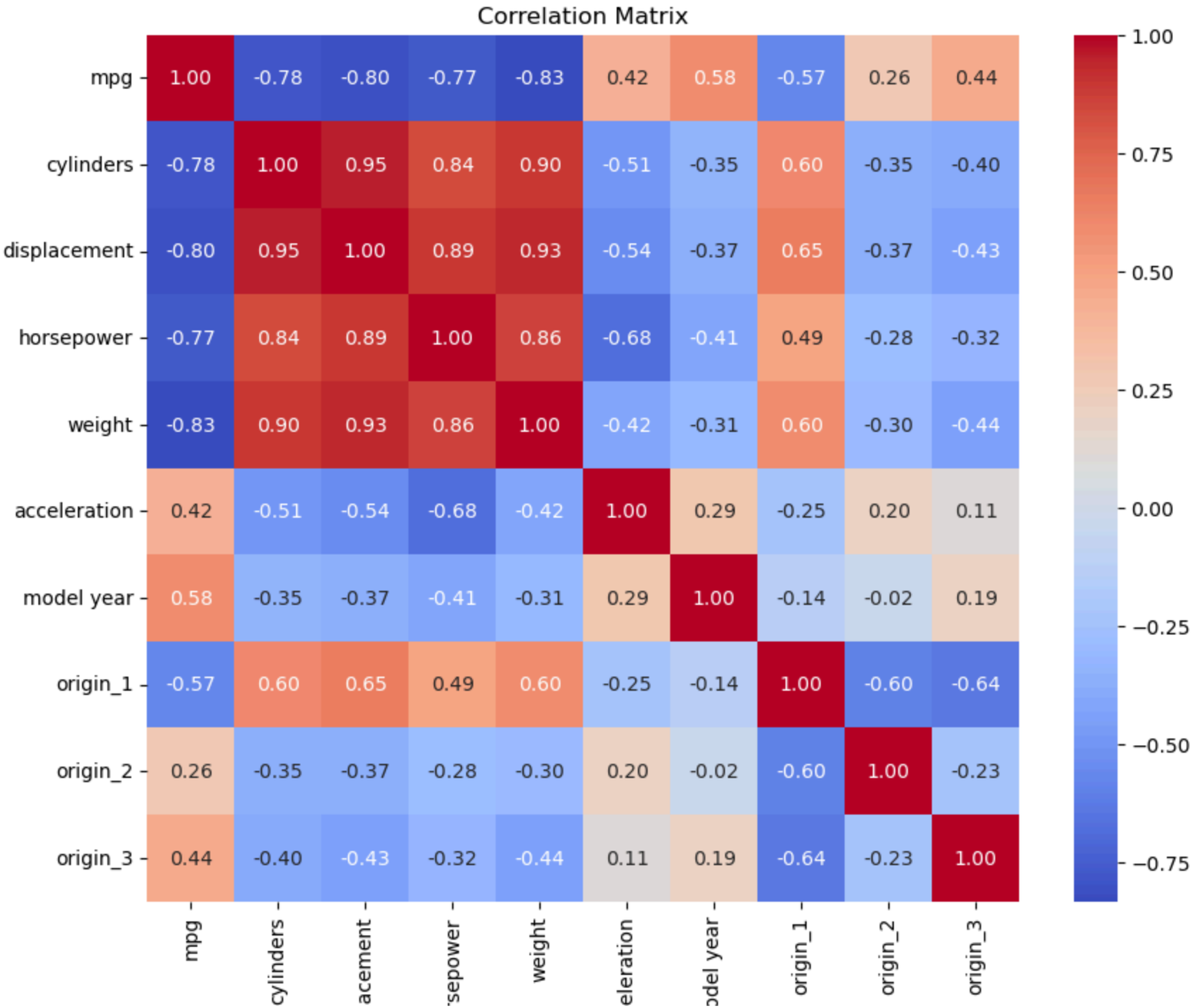
- Remove the car name column.
- The horsepower column values likely imported as a string data type. Figure out why and replace any strings with the column mean.
- Create dummy variables for the origin column.
- Create a correlation coefficient matrix and/or visualization. Are there features highly correlated with mpg?
- Plot mpg versus weight. Analyze this graph and explain how it relates to the corresponding correlation coefficient.

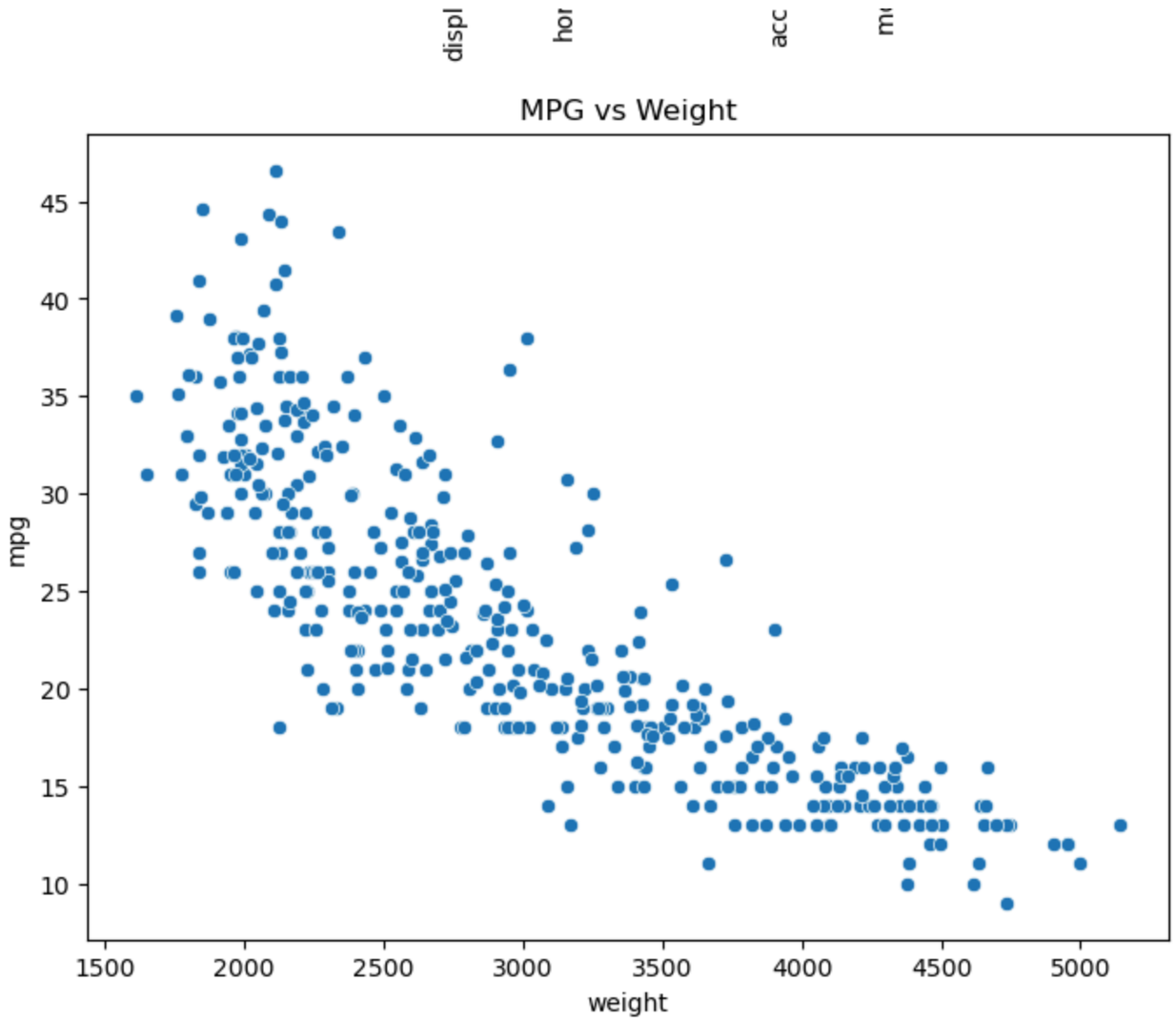
In [31]:

```python
# 2a. Remove the car name column
df.drop('car name', axis=1, inplace=True)

# 2b. Handle horsepower column. It contains "?" which is a non-numeric values, so we will convert it to numeric and
# fill non-numeric values with the mean horsepower.
# Identify non-numeric values
df['horsepower'] = pd.to_numeric(df['horsepower'], errors='coerce')
hp_mean = df['horsepower'].mean()
df['horsepower'] = df['horsepower'].fillna(hp_mean)

# 2c. Create dummy variables for the origin column.
# The origin column (representing regions 1, 2, and 3) was transformed into dummy variables (origin_1, origin_2, origin_3)
# to allow for better analysis and modeling.
df = pd.get_dummies(df, columns=['origin'], prefix='origin')

# 3. Correlation coefficient matrix and visualization
# The correlation matrix was computed to understand the relationships between different variables in the dataset.
# A heatmap was created to visualize these correlations, with annotations showing the correlation coefficients.
# This helps identify which features are most strongly correlated with miles per gallon (mpg)
# and can inform feature selection for modeling.
# Highest Negative Correlations: weight ($-0.83$), displacement ($-0.80$), cylinders ($-0.78$), and horsepower ($-0.77$).
# Highest Positive Correlations: model year ($0.58$) and acceleration ($0.42$).
corr_matrix = df.corr()
plt.figure(figsize=(10, 8))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Matrix')
plt.savefig('correlation_matrix.png')

# 4. Plot mpg versus weight
# A scatter plot was created to visualize the relationship between miles per gallon (mpg) and weight.
# The scatter plot (see below) shows a clear negative non-linear relationship.
# As the weight of the vehicle increases, the miles per gallon (MPG) decreases significantly.
# This visual trend aligns with the strong negative correlation coefficient of $-0.83$,
# confirming that weight is one of the most significant predictors of fuel efficiency.

plt.figure(figsize=(8, 6))
sns.scatterplot(x='weight', y='mpg', data=df)
plt.title('MPG vs Weight')
plt.savefig('mpg_vs_weight.png')
```

```python
print("Correlation with mpg:")
print(corr_matrix['mpg'].sort_values(ascending=False))
```

```
Correlation with mpg:
mpg             1.000000
model year      0.579267
origin_3        0.442174
acceleration    0.420289
origin_2        0.259022
origin_1       -0.568192
horsepower     -0.771437
cylinders      -0.775396
displacement   -0.804203
weight         -0.831741
Name: mpg, dtype: float64
```


Correlation Matrix
mpg
cylinders
displacement
horsepower
weight
acceleration
model year
origin_1
origin_2
origin_3
1.00 -0.78 -0.80 -0.77 -0.83 0.42 0.58 -0.57 0.26 0.44
-0.78 1.00 0.95 0.84 0.90 -0.51 -0.35 0.60 -0.35 -0.40
-0.80 0.95 1.00 0.89 0.93 -0.54 -0.37 0.65 -0.37 -0.43
-0.77 0.84 0.89 1.00 0.86 -0.68 -0.41 0.49 -0.28 -0.32
-0.83 0.90 0.93 0.86 1.00 -0.42 -0.31 0.60 -0.30 -0.44
0.42 -0.51 -0.54 -0.68 -0.42 1.00 0.29 -0.25 0.20 0.11
0.58 -0.35 -0.37 -0.41 -0.31 0.29 1.00 -0.14 -0.02 0.19
-0.57 0.60 0.65 0.49 0.60 -0.25 -0.14 1.00 -0.60 -0.64
0.26 -0.35 -0.37 -0.28 -0.30 0.20 -0.02 -0.60 1.00 -0.23
0.44 -0.40 -0.43 -0.32 -0.44 0.11 0.19 -0.64 -0.23 1.00
mpg
cylinders
acement
sepower
weight
eleration
odel year
origin_1
origin_2
origin_3
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75



- Randomly split the data into 80% training data and 20% test data, where your target is mpg.
- Train an ordinary linear regression on the training data.
- Calculate R2, RMSE, and MAE on both the training and test sets and interpret your results.
- Pick another regression model and repeat the previous two steps. Note: Do NOT choose logistic regression as it is more like a classification model.

In [32]:
```python
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, Ridge
from sklearn.metrics import r2_score, mean_squared_error, mean_absolute_error

# 5. Split the data (80/20)
X = df.drop('mpg', axis=1)
y = df['mpg']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

def evaluate_model(model, X_train, X_test, y_train, y_test, name):
    # Train
    model.fit(X_train, y_train)

    # Predict
    train_preds = model.predict(X_train)
    test_preds = model.predict(X_test)

    # 7. Metrics
    metrics = {
        'R2 Train': r2_score(y_train, train_preds),
        'R2 Test': r2_score(y_test, test_preds),
        'RMSE Train': np.sqrt(mean_squared_error(y_train, train_preds)),
        'RMSE Test': np.sqrt(mean_squared_error(y_test, test_preds)),
        'MAE Train': mean_absolute_error(y_train, train_preds),
        'MAE Test': mean_absolute_error(y_test, test_preds)
    }

    print(f"--- {name} Results ---")
    for k, v in metrics.items():
        print(f"{k}: {v:.4f}")
    return metrics

# 6. Ordinary Linear Regression
lr_model = LinearRegression()
lr_results = evaluate_model(lr_model, X_train, X_test, y_train, y_test, "Linear Regression")

# 8. Another regression model: Ridge Regression
ridge_model = Ridge(alpha=1.0)
ridge_results = evaluate_model(ridge_model, X_train, X_test, y_train, y_test, "Ridge Regression")

# Performance: Both models explain approximately 84.5% of the variance in the test set.
```

```
# Error: The RMSE of 2.89 means that on average, our predictions deviate from the actual MPG by about $2.89$ units.
# The MAE of 2.29 shows the average absolute error is slightly lower.
# Generalization: Interestingly, the models performed slightly better on the test set than the training set
# (higher R^2, lower error).
# This indicates the models are well-generalized and not overfitting to the training data.
# Ridge vs. Linear: The results for Ridge and Linear regression are nearly identical,
# suggesting that the features do not have extreme multicollinearity or large coefficients
# that require heavy regularization in this specific split.
```

```
--- Linear Regression Results ---
R2 Train: 0.8188
R2 Test: 0.8449
RMSE Train: 3.3703
RMSE Test: 2.8878
MAE Train: 2.6055
MAE Test: 2.2876
--- Ridge Regression Results ---
R2 Train: 0.8188
R2 Test: 0.8449
RMSE Train: 3.3703
RMSE Test: 2.8879
MAE Train: 2.6051
MAE Test: 2.2886
```